EXPNO:3	CLASSIFICATION WITH DECISION TREES

AIM

To implement a Decision Tree classifier and evaluate its performance using accuracy score and confusion matrix on a real-world dataset.

ALGORITHM

1. Import necessary libraries
2. Load a classification dataset (e.g., Iris or Titanic)
3. Split the dataset into training and test sets
4. Preprocess data if needed
5. Train a DecisionTreeClassifier from sklearn.tree
6. Predict on test data
7. Evaluate using:
    a. Confusion Matrix
    b. Accuracy Score
8. Visualize the Decision Tree (optional)

CODE:

```
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix, accuracy_score
import matplotlib.pyplot as plt
import seaborn as sns

iris = load_iris()

X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
```

```
)

dt_model = DecisionTreeClassifier(
    criterion="entropy",
    max_depth=3,
    random_state=42
)

dt_model.fit(X_train, y_train)

y_pred = dt_model.predict(X_test)

cm = confusion_matrix(y_test, y_pred)
acc = accuracy_score(y_test, y_pred)

print("Confusion Matrix:")
print(cm)
print("Accuracy Score:", acc)

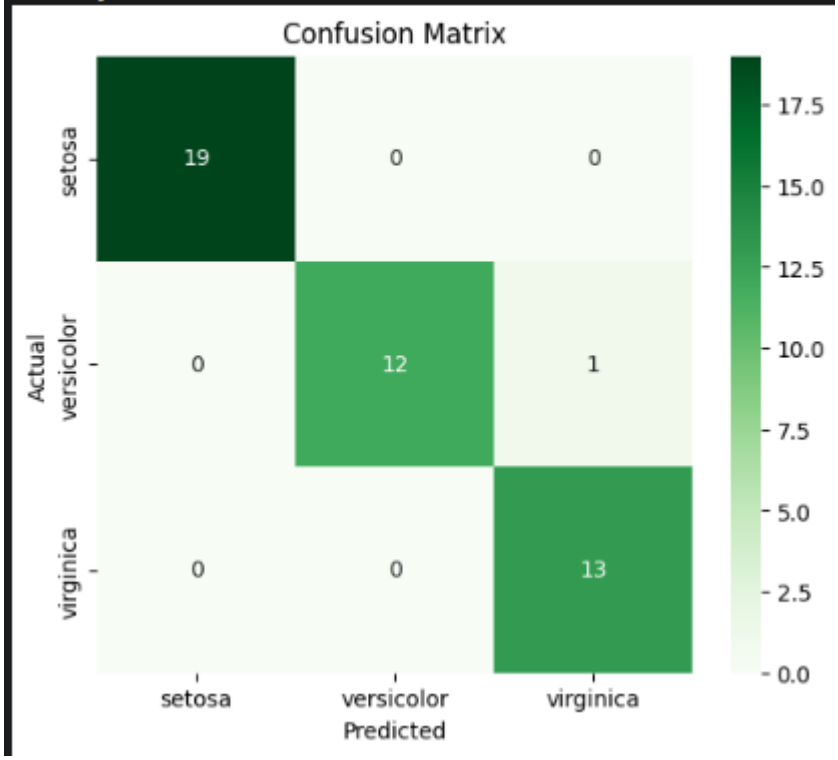
plt.figure(figsize=(6,5))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Greens",
    xticklabels=iris.target_names,
    yticklabels=iris.target_names
)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()

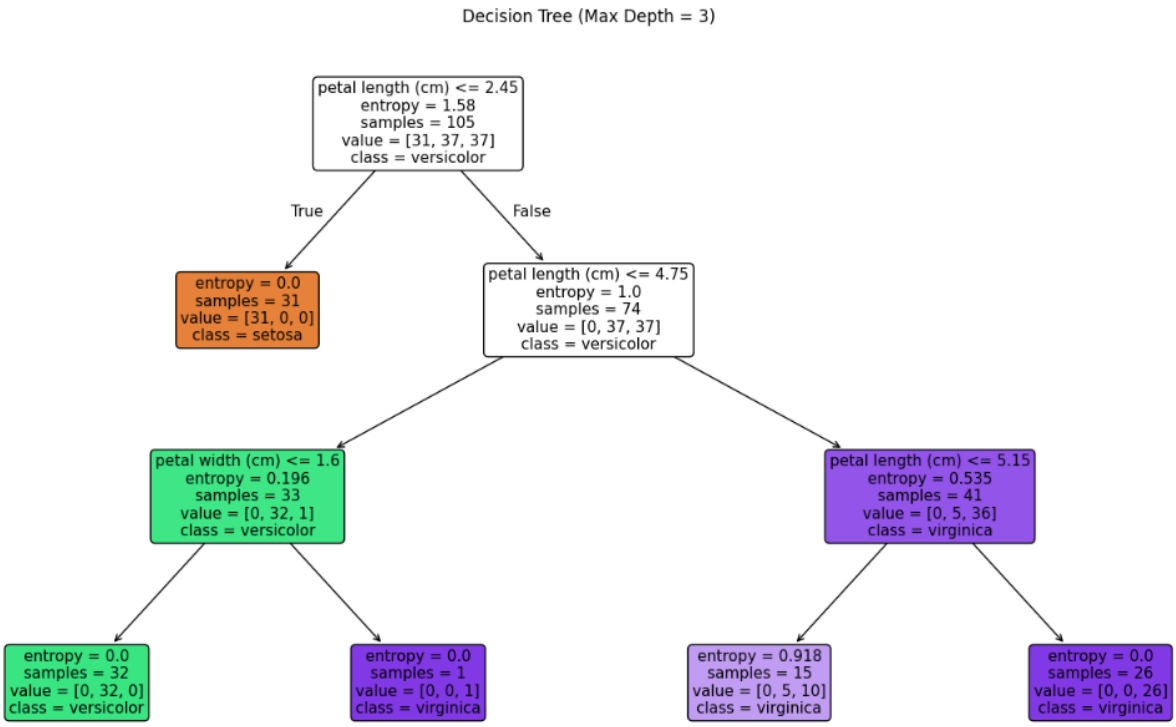
plt.figure(figsize=(14,8))
plot_tree(
    dt_model,
```

```
    feature_names=iris.feature_names,
    class_names=iris.target_names,
    filled=True,
    rounded=True,
    fontsize=11
)
plt.title("Decision Tree (Max Depth = 3)")
plt.tight_layout()
plt.show()
```

Output:

```
Confusion Matrix:
[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]
Accuracy Score: 0.9777777777777777
```





RESULT:

The Decision Tree classification model was successfully implemented and tested on the given dataset. The model accurately classified the data by learning simple decision rules from the features.

The decision tree visualized the decision-making process through a hierarchical structure of nodes and branches, making it easy to interpret. The classification achieved good accuracy, demonstrating that Decision Trees are effective for both categorical and numerical data,

providing clear and interpretable results.